Venkata Lakshmi Parimala Pasupuleti

Senior Backend Software Engineer · Python · Kotlin · Distributed Systems · AWS · MySQL · Kubernetes · Partner & Merchant Systems · High-Scale Finance

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PROFESSIONAL SUMMARY

Senior Backend Software Engineer with 8+ years designing and developing distributed systems at high scale in financial and compliance environments. At Entrata, owned backend architecture for financial settlement, payments integration, partner-facing APIs, commerce platform, and operational workflows serving 100K+ monthly active users at 99.5% uptime — Python, Kotlin/JVM, Java, PostgreSQL, MySQL, RabbitMQ, AWS, Kubernetes. Built reliable, extensible solutions from system design through deployment with strong technical ownership: 20+ production modules, 300+ code reviews, TDD, CI/CD, zero compliance violations. Engages cross-functional stakeholders and global engineering partners with clarity. Also builds AI systems: AgentProbe (open-source eval toolkit) and Leasel (8-agent LangGraph platform, 93.6% precision). M.S. Data Analytics Engineering, George Mason University (GPA 3.87). — github.com/venkatalakshmiparimala/AgentProbe

WORK EXPERIENCE

Blue Horse Digital
Sep 2026 – Present

AI/ML Engineer (Volunteer) · Remote

- Building ML models, data pipelines, and production APIs; integrating AI components across backend layers; debugging reliability issues; contributing to team growth through development activities and proactive feedback.

George Mason University
Aug 2024 – Aug 2025

Research Associate + Teaching Assistant

- Built multi-source ML prediction pipelines (Python, R) across 190+ countries (R^2=0.977); HuggingFace fine-tuning + inference API deployment; statistical modeling and validation. Workshops to 600+ students across Python and SQL.

Entrata (PropTech / Real Estate SaaS)
Sep 2018 – Aug 2024

Senior Software Engineer · Remote

- **Backend system design and development at high scale** (Python, Kotlin/JVM, Java, PostgreSQL, MySQL, RabbitMQ, AWS, Kubernetes) — owned 20+ production modules serving 100K+ monthly active users at 99.5% uptime across 120+ Agile sprints. Designed and maintained partner-facing APIs, merchant interface workflows, financial settlement, payments integration, and operational workflows in a high-scale financial environment where correctness is non-negotiable. Zero compliance violations. Zero security incidents.
- **Reliable, extensible distributed systems** — event-driven architecture (RabbitMQ) with retry logic, idempotency guarantees, and fault-tolerant design; multi-tenant PostgreSQL and MySQL at scale; Docker microservices redesign (50% scalability improvement, 40% deployment time reduction); 200+ SQL optimizations (30% performance gain); AWS (S3, EC2, RDS, Lambda, Step Functions); Kubernetes orchestration. Balanced speed and quality to meet business objectives while delivering extensible solutions designed for long-term evolution.
- **Technical ownership, code quality, and global collaboration** — 300+ peer code reviews; TDD and well-tested, maintainable code across all modules; CI/CD (GitHub Actions) with automated testing; debugged complex codebases and resolved production defects under SLA pressure; OpenTelemetry + Prometheus + Grafana (40% debugging time reduction). Engaged cross-functional stakeholders to improve visibility and communication; collaborated effectively with global engineering partners. Mentored 5+ engineers; proactive approach to feedback and personal growth.

Entrata India, Pune
Jul 2017 – Sep 2018

Associate Software Engineer

- Built financial and compliance backend modules (PHP, Python, Java, PostgreSQL, MySQL); led AWS S3 migration; data pipelines across hundreds of thousands of financial and partner records.

PROJECTS

AgentProbe: Open-Source AI Evaluation Toolkit
CI-INTEGRATED · OPEN SOURCE

Python · FastAPI · LangGraph · ChromaDB · GitHub Actions | github.com/venkatalakshmiparimala/AgentProbe

- 15-endpoint FastAPI backend; 5-metric eval engine; LLM-as-judge scoring; CI regression harness; distributed model routing across OpenAI/Anthropic/Groq/Ollama. High-quality, maintainable, extensible code built with production Python engineering practices and full test coverage.

Leasel: 8-Agent AI Compliance Platform
LANGGRAPH · FINANCIAL COMPLIANCE

Python · LangGraph · FastAPI · SQLite · OpenAI API | github.com/venkatalakshmiparimala/Lease_complaince_rag

- Distributed 8-agent workflow with durable state, fault-tolerant HITL escalation, and RAG over financial compliance documents. 93.6% precision, 87.9% F1 across 1,240 compliance test cases. Core design: deterministic logic governs every regulated financial outcome.

EDUCATION

M.S. Data Analytics Engineering George Mason University, Virginia — May 2026

GPA: 3.87

B.Tech Computer Science & Engineering JNTU Anantapur — 2017

SKILLS

Languages & Frameworks: Python (8+ yrs), Kotlin (JVM), Java (Spring Boot), TypeScript, JavaScript, Node.js, PHP, SQL, PL/SQL, Bash — FastAPI, Django, React.js, REST APIs, GraphQL

Backend & Distributed Systems: PostgreSQL, MySQL, Redis, RabbitMQ, Apache Kafka, Temporal, Microservices, Distributed Systems, System Design, Event-Driven Architecture, Partner APIs, Merchant Interface, Operational Workflows, Multi-Tenant Architecture, Idempotency, Fault-Tolerant Systems, Docker, Kubernetes (K8s), AWS (S3, EC2, RDS, Lambda, Step Functions), Terraform, GitHub Actions CI/CD, TDD, Unit Testing, Integration Testing, Extensible Code

Financial & Observability: Financial Settlement, Payments Integration, Commerce Platform, PHI Compliance, HIPAA, SOC 2, Audit Logging, OpenTelemetry, Prometheus, Grafana, Performance Engineering, LangGraph, RAG Pipelines, LLM Evaluation, Agile/Scrum